

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL2 - CONCEPT MAPPING

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	C01	Assessment:	ASSESSMENT TOOL 2- CONCEPT MAPPING
Weightage:	35%	Total Marks:	25
C01 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
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Department:	IT	Date:	14-7-26

Instruction to Students

1. Answer two questions as per the Instruction.
2. Each question carries 12.5 mark.
3. Only the appropriate Tools can be used
4. Time allotted for the exam is 60 min

ANNEXURE A

1. Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance.
2. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.
3. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.
4. Draw a detailed concept map that connects various addressing modes (immediate, direct, register, indirect, indexed) with instruction types (data transfer, arithmetic, control instructions). Illustrate how each addressing mode affects memory access time, instruction size, and execution speed, and show real-world scenarios where specific addressing modes are preferred.
5. Prepare an elaborate concept map integrating number systems (binary and hexadecimal) with instruction representation, memory storage, and debugging processes. Show how data is

converted between formats, how instructions are stored and interpreted by the CPU, and why hexadecimal representation is commonly used in low-level programming and system design.

Code of Conduct:

I Logesh P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: REVANTHSAI

MARKS ALLOCATION

Criteria	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)
Q1					
Q2					

RUBRICS FOR CALCULATION:

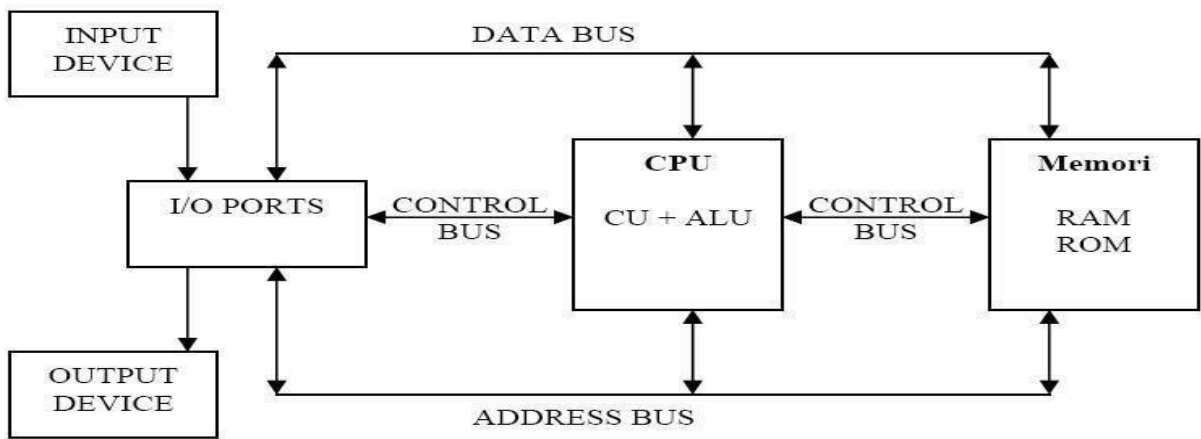
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are wellstructured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence

Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand
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Question 1: CPU, Memory, I/O Devices and Bus Structures, Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance

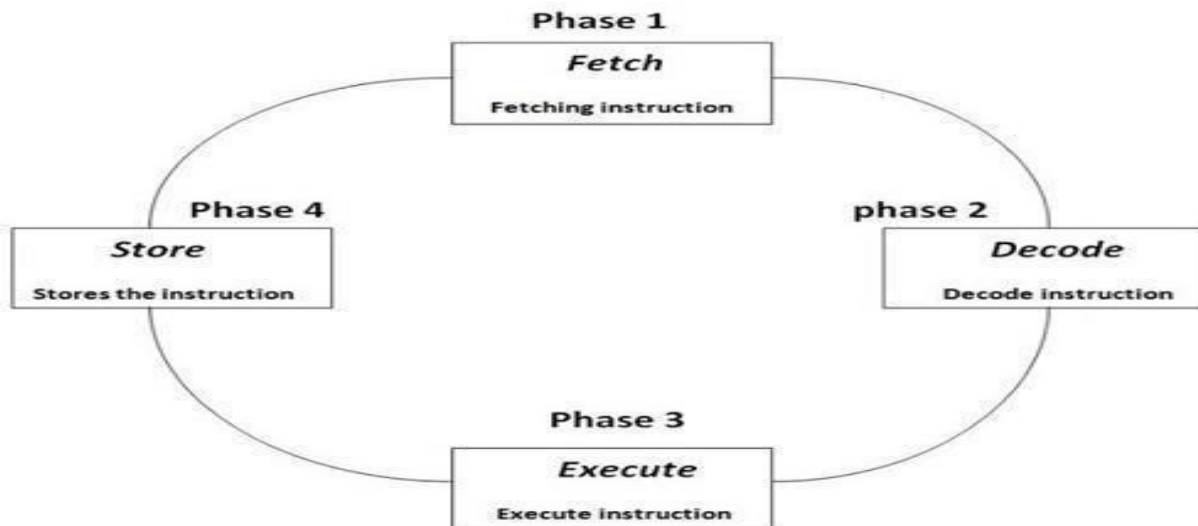
Answer

Flowchart (Image):

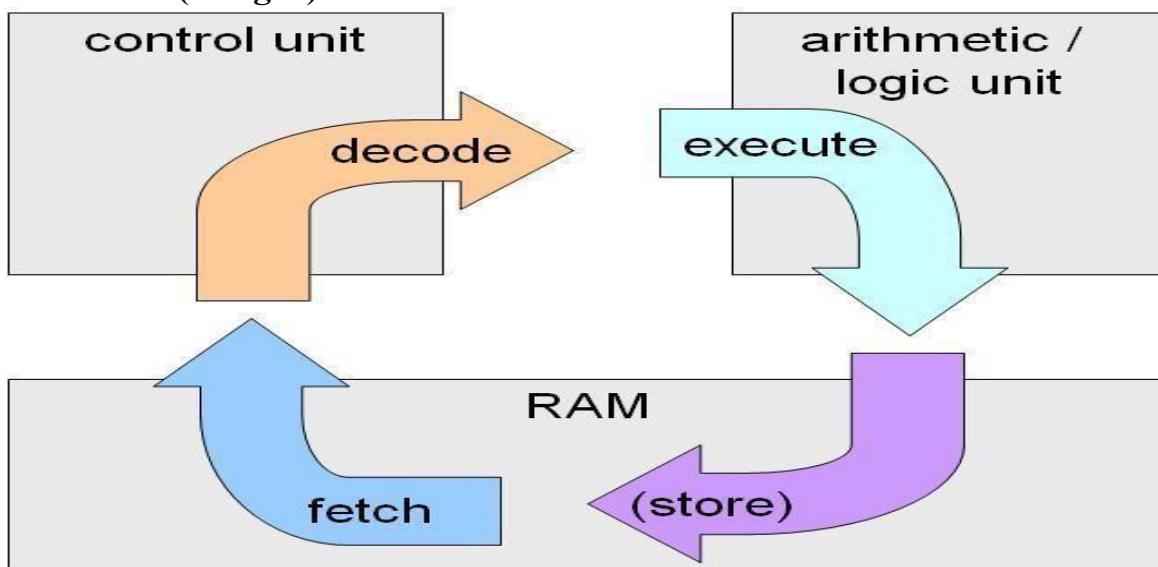


Question 2: Instruction Execution Cycle and CPU Performance Answer

Flowchart (Image):



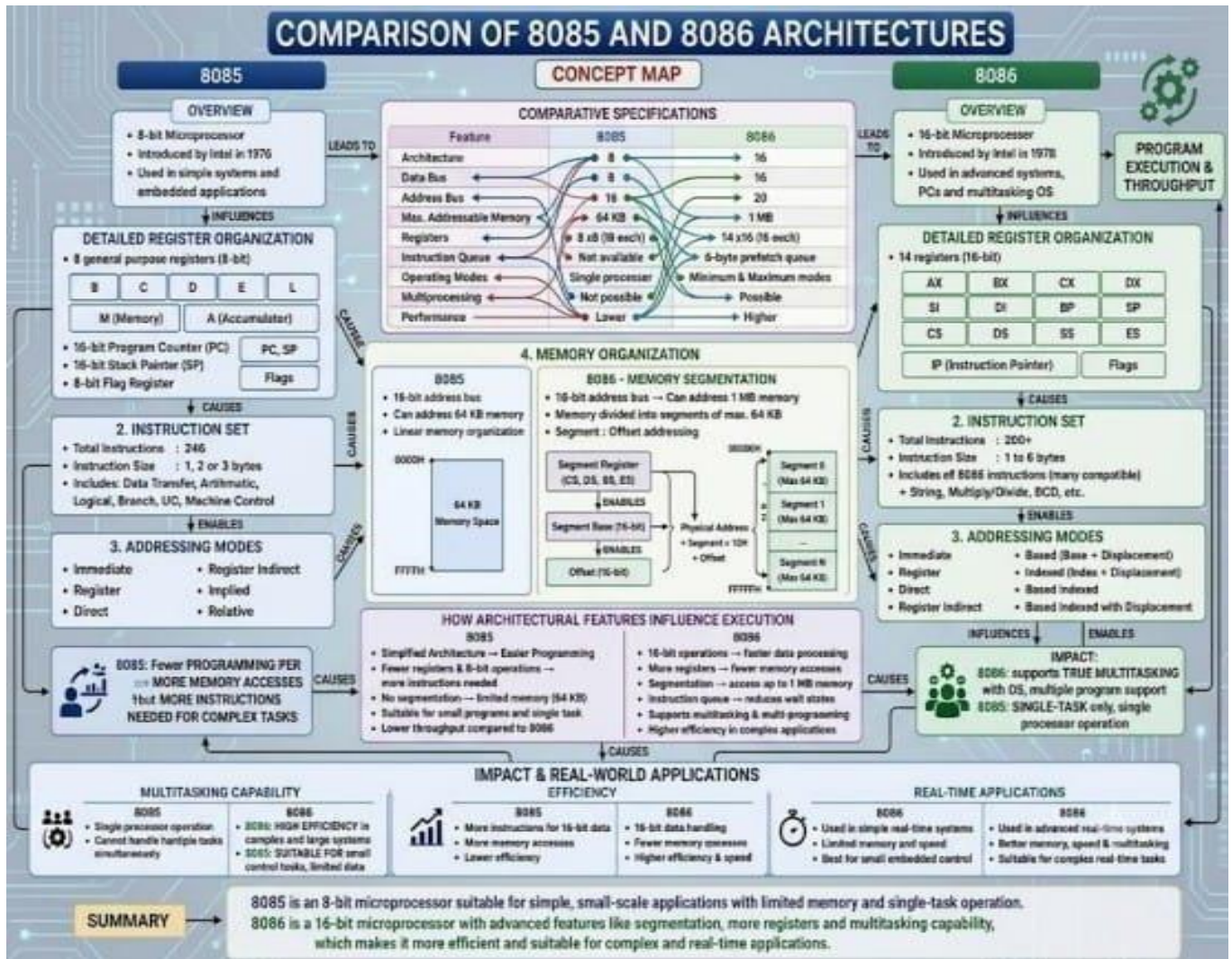
Flowchart (Image2):



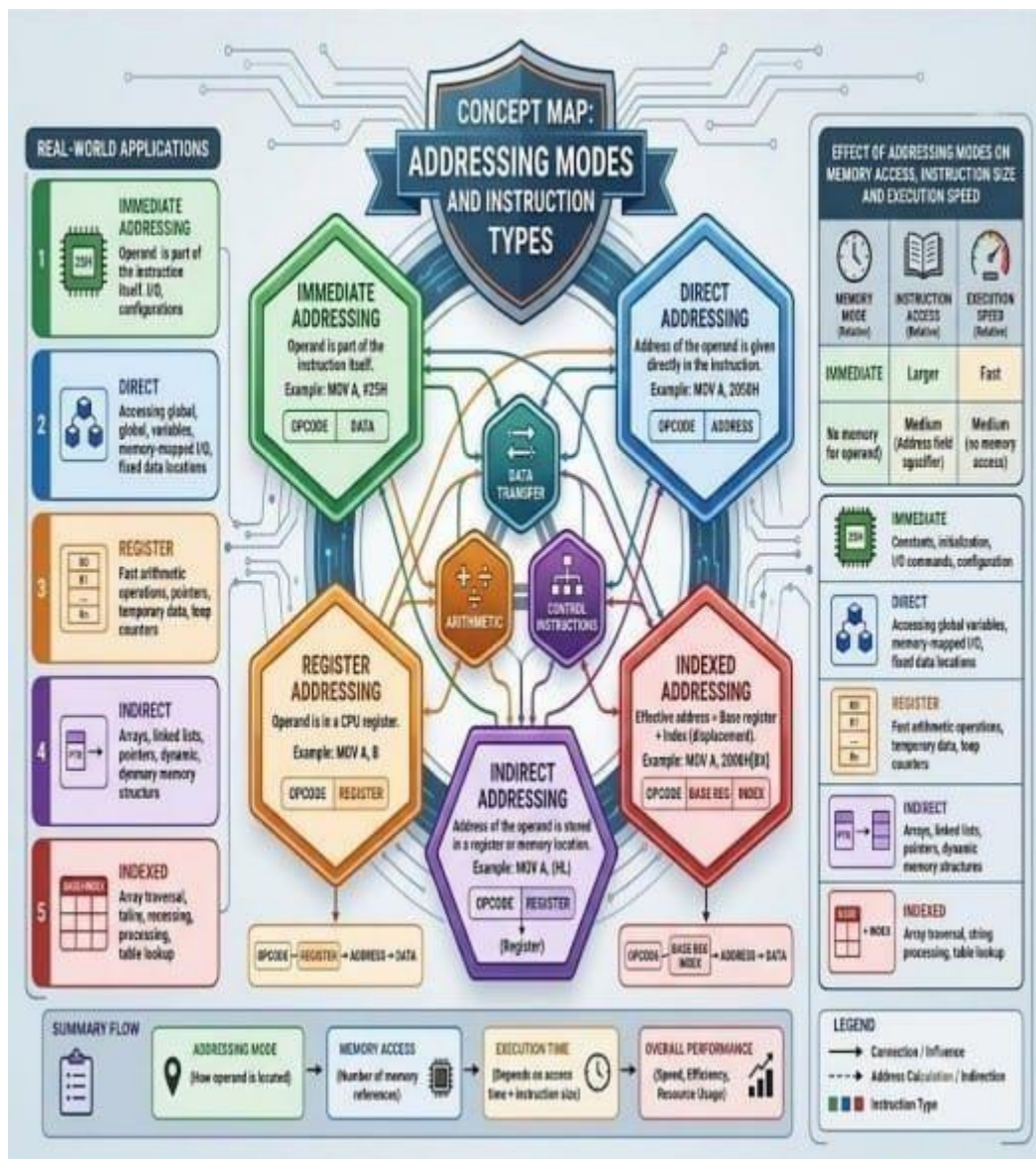
Conclusion:

The Instruction Execution Cycle (Fetch–Decode–Execute) forms the foundation of CPU operation, with each stage directly influencing overall system performance. The fetch stage depends on efficient memory access, the decode stage is affected by instruction complexity, and the execute stage determines how quickly operations are completed. These factors collectively impact Clock Cycles Per Instruction (CPI), the total clock cycles, and ultimately the CPU execution time. Pipeline stalls, cache misses, and complex instructions can increase CPI and reduce performance, while optimized pipelines and faster memory improve efficiency. Benchmarking plays a crucial role by measuring and comparing CPU performance under standardized workloads, helping evaluate how effectively a processor executes instructions in realworld scenarios.

3. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.

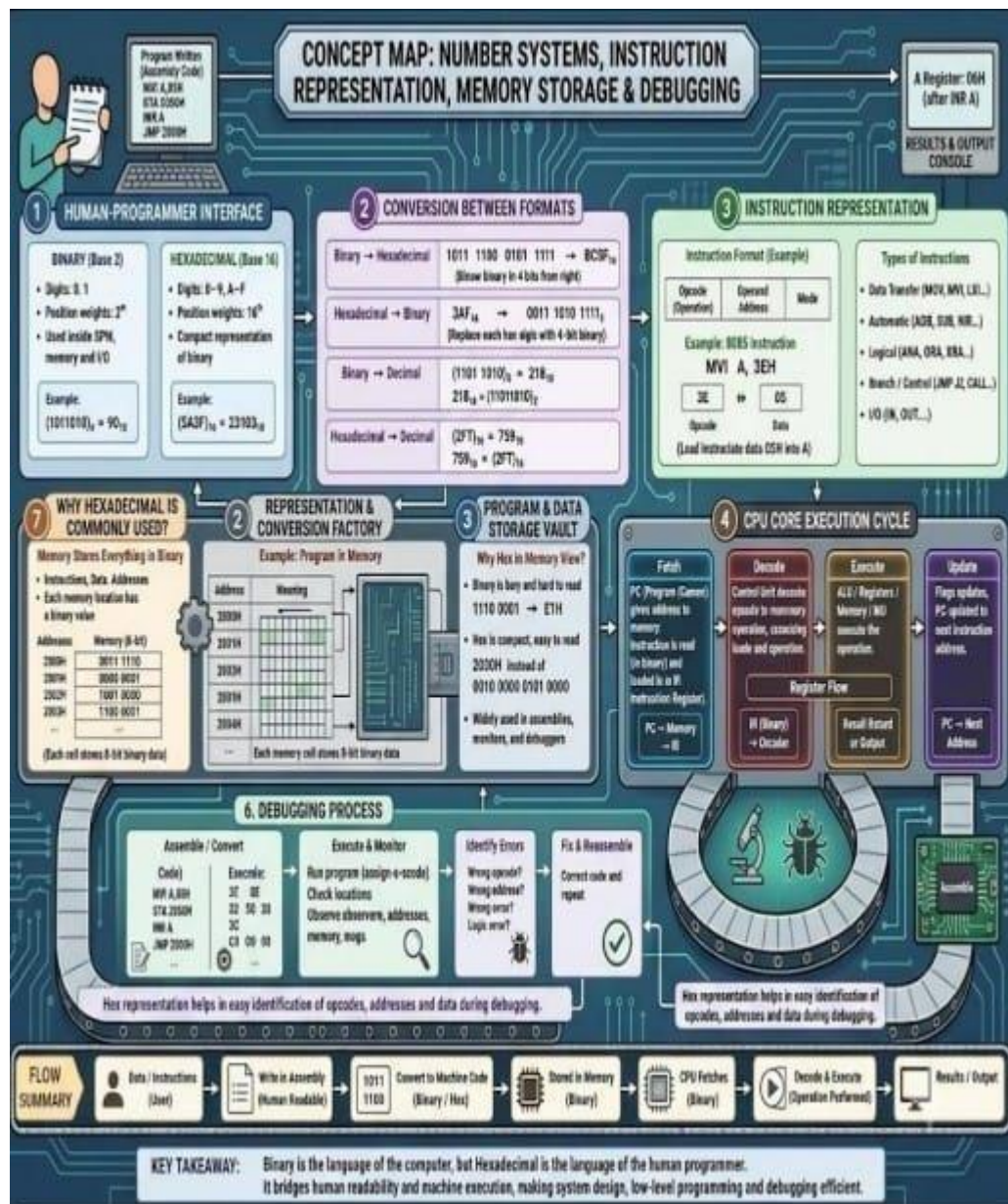


4. Draw a detailed concept map that connects various addressing modes (immediate, direct, register, indirect, indexed) with instruction types (data transfer, arithmetic, control instructions). Illustrate how each addressing mode affects memory access time, instruction size, and execution speed, and show realworld scenarios where specific addressing modes are preferred.



5.Prepare an elaborate concept map integrating number systems (binary and hexadecimal) with instruction representation, memory storage, and debugging processes. Show how data is converted between formats, how instructions are stored and interpreted by the CPU, and why hexadecimal representation is commonly used in low-level programming and system design.

Answer:



6. DEBUGGING PROCESS

Assemble / Convert

Code	Assembler
MOV A, R0H	3E 0E
STA 2050H	32 50 30
INR A	3C
JMP 2050H	C3 00 00

Execute & Monitor

Run program (assign e-serials). Check locations. Observe observance, addresses, memory maps.

Identify Errors

Wrong operand? Wrong address? Wrong serial? Logic error?

Fix & Reassemble

Correct code and repeat.

Hex representation helps in easy identification of opcodes, addresses and data during debugging.

FLOW SUMMARY

1 Data / Instructions (User)

2 Write in Assembly (Human Readable)

3 1011 1100

4 Convert to Machine Code (Binary / Hex)

5 Stored in Memory (Binary)

6 CPU Fetches (Binary)

7 Decode & Execute (Operation Performed)

8 Results / Output

KEY TAKEAWAY: Binary is the language of the computer, but Hexadecimal is the language of the human programmer. It bridges human readability and machine execution, making system design, low-level programming and debugging efficient.